Local pizza competition and menu prices

Evidence from the Houston area

Econometrics project Group 9

Economic question

Do restaurants facing greater local pizza competition charge lower average pizza prices?

Expected mechanism

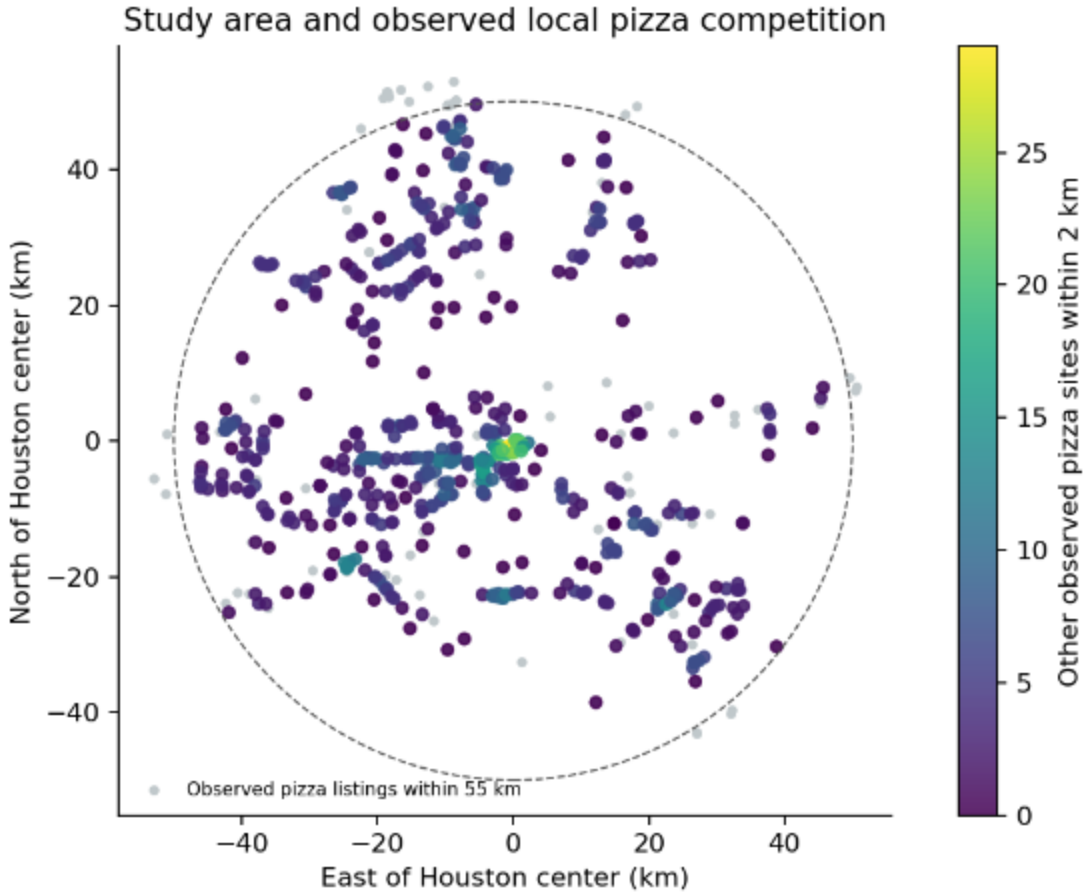
Nearby substitutes may constrain restaurant markups.

Location also matters

Demand and rents may raise both restaurant density and prices.

We compare observed listings, holding included characteristics fixed.

Houston study area



597

listings with usable prices

50 km focal radius
55 km competitor buffer

704 observed supplier sites in the buffered area

Same original Uber Eats dataset. Source files dated November 2023.

Construction of the restaurant price

Mean listed price of retained whole-pizza offers

Product rules

Positive USD prices
Whole pizzas
8–18 inches when size is explicit
Duplicate eligible offers removed

Product comparability

Sizes are often missing.
Mean known-size share: 21.2%.

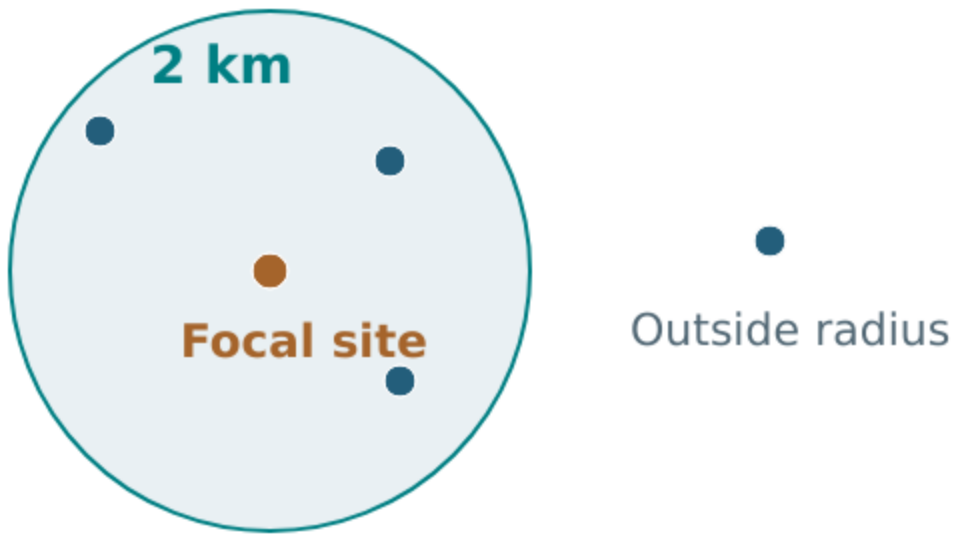
Menu averages still mix products.

Unknown sizes remain. Each retained menu offer receives equal weight.

Local competition within 2 km

Distinct sites, excluding the entire own site

Illustration: 3 nearby rival sites



5.17

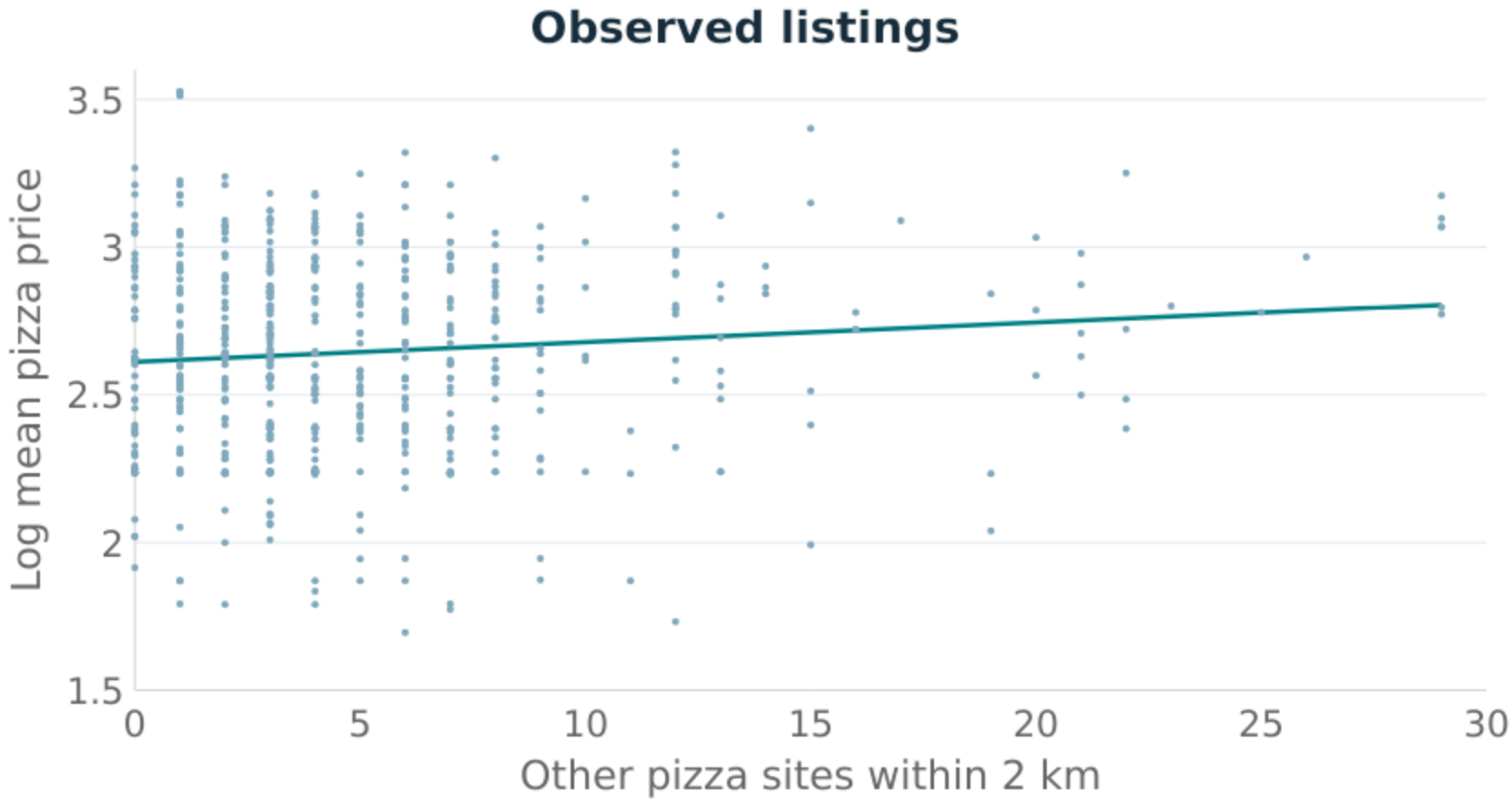
sites on average

0–29

observed range

Great-circle distances. Suppliers with missing prices or ratings remain in the count.

The raw relationship explains little price variation



1.1%

of log-price variation explained by M1

597 listings. The fitted line describes an unadjusted association.

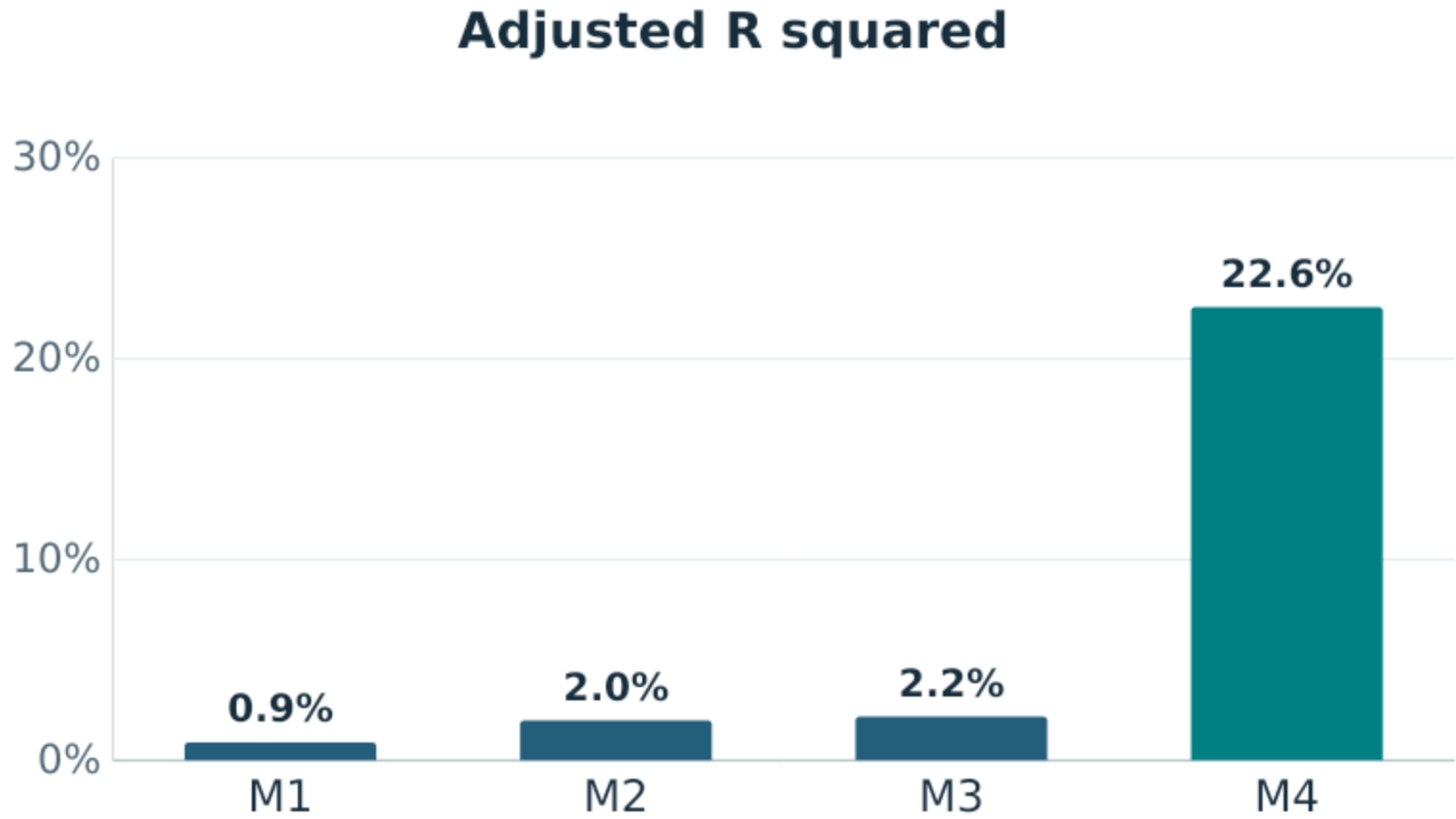
Model sequence on the same 597 listings

Model	Regressors	Purpose
M1	Nearby competitor count	Raw association
M2	M1 + location controls	Spatial price differences
M3	M2 + identified-chain indicator	Observable brand grouping
M4	**M3 + menu-composition controls**	**More comparable offers**

log(Mean pizza price) = α + β Competition + controls + u

M4 is the preferred specification. Main coefficient tables use HC3 standard errors.

Model fit improves with menu controls



Holdout dollar RMSE

M1 $4.62

M4 $3.91

477 training listings
120 holdout listings

One fixed internal split. Nearby and related listings can appear in both sets.

Main result

+0.59%

per additional observed pizza site
within 2 km, holding controls fixed

Uncertainty estimate	95% interval	p value
HC3	+0.12% to +1.05%	0.0129

Log-price coefficient transformed with 100 × (exp(β) − 1). Conditional association.

One extension: chain differences

M5: log(Price) = … + β Competition + θ Chain + δ Competition × Chain

Listing group	Slope per site	HC3 95% interval
Other or unclassified	+0.70%	+0.14% to +1.27%
Identified chains	+0.30%	−0.58% to +1.19%

Difference in slopes p = 0.450

The identified-chain slope is β + δ. Group differences use the full covariance.

Four tests, four explicit questions

Test	Question	p value
T1 · M4	Is the competition coefficient zero?	0.0129
T2 · M4	Are all 11 control coefficients zero?	< 0.0001
T3 · M4	Are all 12 slopes zero?	< 0.0001
T4 · M5	Is the chain slope difference zero?	0.450

These tests do not establish a causal effect.

T1: two-sided t test. T2–T4: HC3 Wald F tests. Nominal 5% threshold.

What the model still misses

Check	Result	Implication
RESET for M4	$p = 0.0058$	Some price patterns remain unexplained
Breusch–Pagan	$p < 0.0001$	Error variance is not constant

Entry and prices respond to local demand and rents.

Unknown sizes and platform selection limit comparability.

Robust standard errors address uncertainty under assumptions. Endogeneity remains.

What the data support

The preferred model gives a small positive price association.

The expected negative relationship does not emerge in this observed cross-section.

Causal interpretation would require variation that separates competition from local demand.

What the controls contain

Block	Variables
Location	East, north, east², north², east × north
Chain status	One identified-chain indicator
Menu	Mean known diameter, missing-diameter indicator, known-diameter share, small-label share, large-label share
M5 only	Competition × identified-chain indicator

All models use the same 597 listings. M5 extends M4.

Data and reproducible results

Uber Eats USA Restaurants and Menus

Ahmed Shahriar Sakib
Source files dated 22 November 2023

kaggle.com/datasets/ahmedshahriarsakib/uber-eats-usa-restaurants-menus

Group 9 project package

One report, one presentation, the data,
Python code and the editable LaTeX source